

Nagizaaz Shaik

AI / LLM Engineer · ML Systems · MLOps · Cloud Architect

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SUMMARY

5+ years of production experience building LLM systems, multi-agent pipelines, and the MLOps infrastructure that ships them — spanning AI/LLM Engineering, ML Systems, and Cloud Architecture. At Wright-Patt Credit Union, designed and operates the full ML platform — five SageMaker-hosted production models, XGBoost underwriting scoring, a fine-tuned BERT classifier, RAG compliance tooling, and automated retraining — with zero findings across two consecutive compliance audits on \$400M in member financial data. Open-source portfolio of 12 production-grade repos (CI-green, tested) across LLM infrastructure (multi-provider gateways, LLMops evaluation), agentic AI (autonomous coding & SRE agents, multi-agent financial research), and LLM safety (OWASP LLM Top 10 red-teaming). AWS ML Specialty certified.

TECHNICAL SKILLS

LLM & Agentic AI: RAG Pipelines (Hybrid Search · Reranking · HyDE · Corrective RAG) · Multi-Agent Orchestration (LangGraph · CrewAI) · Model Context Protocol (MCP) · Prompt Engineering & Versioning · Semantic Caching · LLM Gateways · OWASP LLM Top 10 · Red-Teaming

LLMOps & Eval: RAGAS · LLM-as-Judge · A/B Testing · Cost Analytics · Agent Observability · MLflow · NVIDIA NIM · vLLM / Ollama

ML & AI: PyTorch · Transformers (from-scratch) · XGBoost · BERT · HuggingFace · scikit-learn · SHAP · AWS SageMaker · SageMaker Model Monitor

Cloud & Infra: AWS (SageMaker · EC2 · S3 · RDS · Glue · Redshift · EKS · IAM · CloudWatch · CloudTrail) · GCP (BigQuery · Vertex AI) · Terraform · AES-256 · RBAC · ECOA

MLOps / DevSecOps: Apache Airflow · MLflow Registry · SageMaker Pipelines · Docker · Kubernetes · ArgoCD (GitOps) · OPA/Conftest · Checkov · GitLeaks · SLSA · FastAPI · CI/CD · Automated Retraining

Data Engineering: Apache Kafka · Spark / PySpark · dbt · AWS Glue · Redshift · Snowflake · Oracle SQL / PL/SQL

Languages: Python · SQL · PL/SQL · Bash

PROFESSIONAL EXPERIENCE

Cloud Architect at Wright-Patt Credit Union · Fairborn, OH

Jul 2024 – Present

- Designed the complete AWS ML platform architecture from scratch — SageMaker inference endpoints, MLflow model registry with version-gated promotion and automated rollback, VPC-isolated compute per environment, and IAM policies scoped to least privilege per service. Deployment failures reduced by 85% within 60 days.
- Defined and enforced the data security architecture across all ML workloads — AES-256 encryption at rest and in transit, CloudTrail logging on all inference calls, and VPC-isolated SageMaker endpoints. Passed two consecutive NCUA compliance audits with zero findings on \$400M in member financial data.
- Deployed an XGBoost credit underwriting model on SageMaker processing 500+ automated loan decisions per day — engineered the complete pipeline from AWS Glue feature preparation to real-time inference endpoint, with S3-backed structured logging for regulatory audit trail. Cut manual analyst review by 30%.
- Built production model monitoring using SageMaker Model Monitor with custom statistical baselines — daily scheduled jobs tracking feature distribution shifts and prediction confidence degradation across five live models. Intercepted two critical drift events before accuracy dropped measurably in production decisions.
- Replaced manual weekly model refresh with an Airflow DAG that retrains on a rolling data window, evaluates against holdout performance, and promotes via SageMaker Pipelines only when AUC, precision, and calibration stay within defined thresholds — eliminating 8+ hours of manual work per week.
- Fine-tuned BERT on internal support ticket data and deployed it as a FastAPI microservice on EC2 — 14 intent categories, sub-200ms p99 latency under production load, routing 10,000+ weekly member requests with a 40% reduction in misrouting rate.
- Built a RAG-based document QA system using LangChain and Pinecone, indexing 500+ internal policy documents and NCUA regulatory guidance — compliance analysts reduced average research time from over two hours to under 20 minutes per query.
- Mentored two engineers transitioning into ML — weekly code reviews and architecture sessions covering model evaluation, deployment patterns, and MLOps fundamentals. Both now independently own and ship production ML components.

Data Engineer at Cognizant · Hyderabad, India

Jul 2020 – Jun 2023

- Led the Oracle 11g to Amazon Redshift migration for a financial services client — redesigned schema for columnar storage with optimal distribution and sort keys, rewrote 20+ stored procedures as modular Airflow DAGs. Average analytical query runtime improved by 60% across a 25GB+ warehouse serving five business units.
- Designed fifteen production ETL/ELT pipelines in Apache Airflow across Oracle, SQL Server, flat files, and REST APIs, with a multi-stage validation layer running row-count reconciliation and null-rate checks at each hop. Surfaced a silent upstream corruption issue that had been invalidating three months of downstream financial reports.
- Built a real-time Kafka feature pipeline for a retail demand forecasting model — POS events aggregated in a five-minute tumbling window and written to Redshift for inference. End-to-end latency under 60 seconds contributed to a 28% reduction in emergency restocking orders.
- Promoted to technical lead within two years — code reviews for three engineers, weekly data modelling sessions, and primary client contact across two accounts with 100% on-time delivery.

SQL / Data Intern at Uber · Hyderabad, India

Jan 2019 – Mar 2019

- Return engagement on the city-ops team — built SQL queries and Vertica views for weekly driver-utilisation, trip-completion, and surge-pricing reporting across Hyderabad.

- Consolidated fragmented ad-hoc queries into standardised templates, cutting analyst prep time for the weekly ops review from 3 hours to under 45 minutes.

Python Intern at Uber · Hyderabad, India

Aug 2018 – Oct 2018

- Automated extraction and transformation of operational event data with Python and pandas — turned a 4-hour manual weekly process into a scheduled job producing clean summary tables for city ops.
- Built an anomaly-detection script flagging unusual trip counts and revenue against rolling 7-day baselines before data entered the reporting pipeline; secured a return offer.

PROJECTS

LLM Gateway **LLM INFRASTRUCTURE**

github.com/shaikn6/llm-gateway

Production LLM gateway with an OpenAI-compatible API — semantic caching, latency-aware routing across Claude / OpenAI / Ollama, token-budget enforcement, and per-key cost analytics behind a single FastAPI interface with Prometheus metrics.

FastAPI · Claude · OpenAI · Ollama · Redis · Prometheus · Docker

LLM Safety Auditor **AI SAFETY**

github.com/shaikn6/llm-safety-auditor

Automated red-teaming framework for production LLMs — 250+ adversarial attack vectors across six mutation strategies (jailbreak, indirect prompt injection, encoding obfuscation, multi-turn escalation). Scores against the full OWASP LLM Top 10 and generates compliance-grade PDF audit reports.

Python · HuggingFace · OWASP LLM Top 10 · Red-Teaming · FastAPI · ReportLab

Finance Agent Crew **AGENTIC AI · FINTECH**

github.com/shaikn6/finance-agent-crew

Multi-agent financial research system — parallel analyst, risk-assessor, and report-writer agents collaborate over SEC filings, earnings calls, and news to produce structured investment briefs with citation chains, orchestrated in LangGraph with n8n automation.

LangGraph · Claude · SEC EDGAR API · FAISS · n8n · FastAPI

Ops Autopilot **AGENTIC AI · SRE**

github.com/shaikn6/ops-autopilot

Autonomous SRE agent — on a Prometheus / Kubernetes alert it performs LLM root-cause analysis, opens a GitHub fix PR, posts a Slack summary, and writes a runbook entry, closing the incident loop without a human paging in.

LangGraph · Anthropic · Prometheus · Kubernetes · PyGithub · Slack API

CERTIFICATIONS

AWS Certified Machine Learning – Specialty · Amazon Web Services

AWS Certified Cloud Practitioner · Amazon Web Services

GCP Professional Data Engineer · Google Cloud

Snowflake SnowPro Advanced: Data Engineer · Snowflake

dbt Analytics Engineering Certification · dbt Labs

EDUCATION

Master of Science, Computer Science · GPA 3.93 / 4.0 — University of Dayton, Dayton, OH

Aug 2023 – May 2025

Coursework: Advanced Machine Learning · NLP · Human-AI Interaction · Big Data Systems · Cloud Computing · Distributed Systems · Statistical Inference

Bachelor of Technology, Computer Science Engineering · GPA 8.9 / 10 — CMR Engineering College, Hyderabad, India

Aug 2016 – May 2020

Coursework: Data Structures & Algorithms · DBMS · OOP (Java) · Computer Networks · Operating Systems · Software Engineering · Design & Analysis of Algorithms